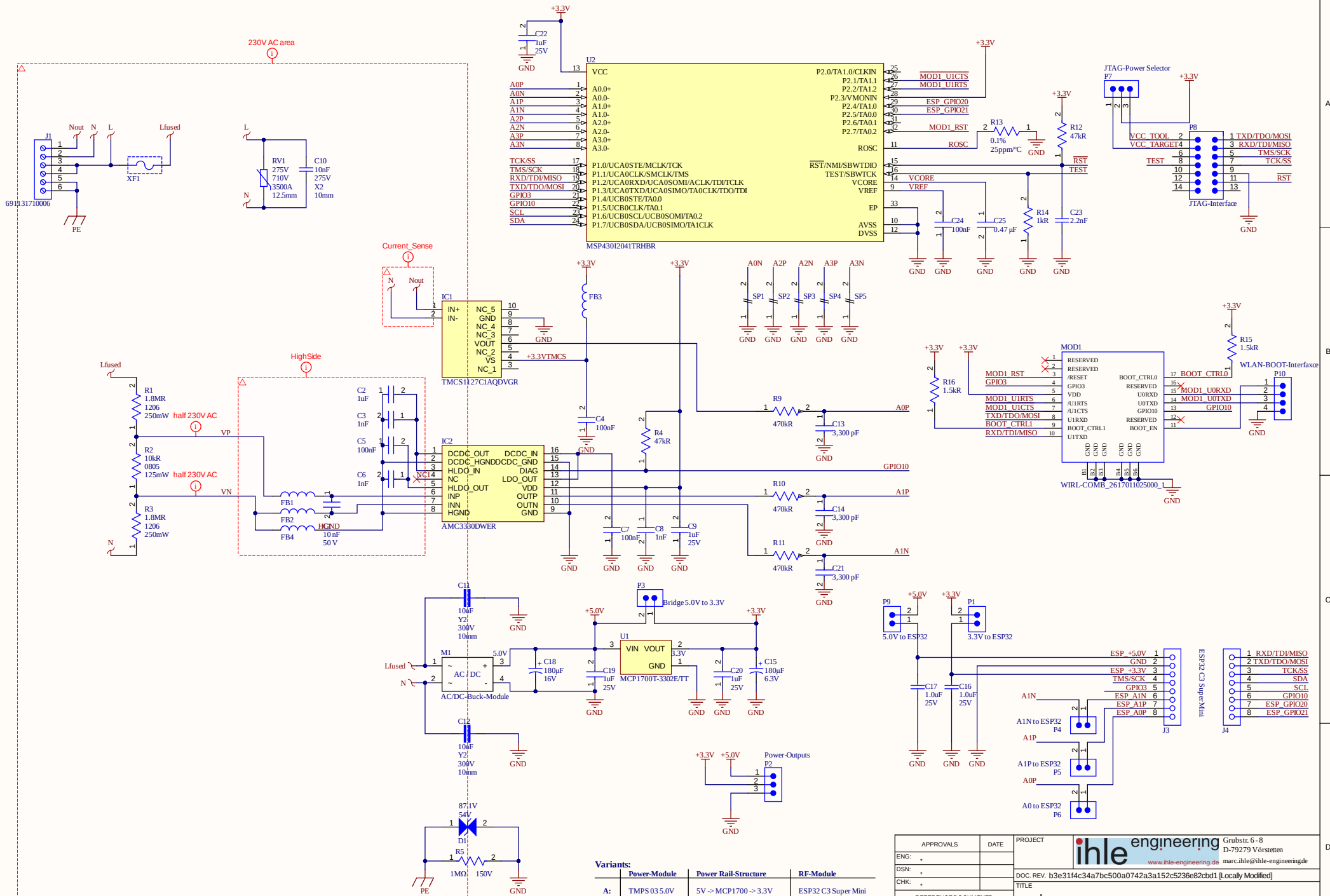


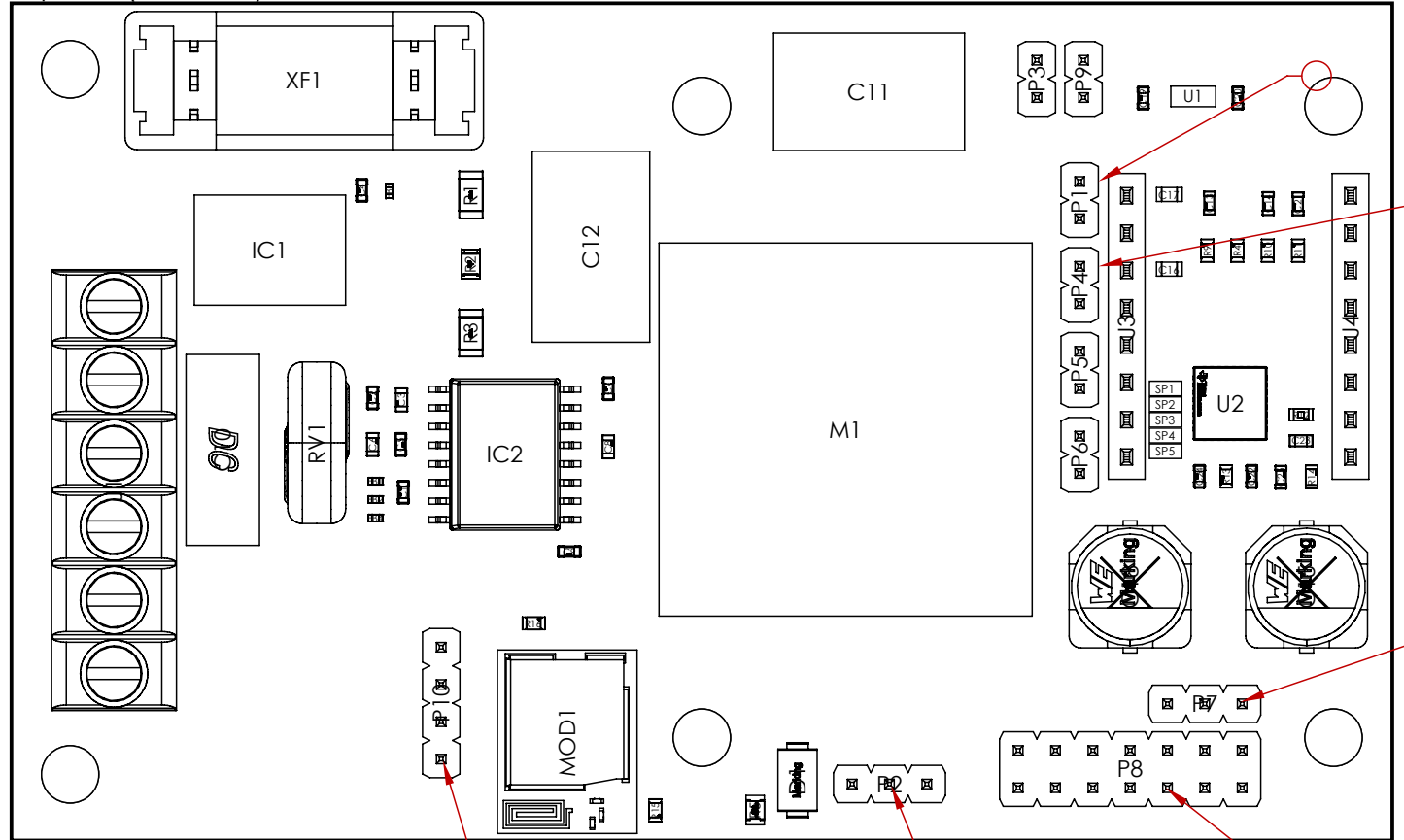


Variants:

	Power-Module	Power Rail-Structure	RF-Module
A:	TMPS03 5.0V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
B:	AM11-12W05V	5V → MCP1700 → 3.3V	ESP32 C3 Super Mini
C:	TMPS03 3.3V	5V = 3.3V (P3 set)	Stephano-I

APPROVALS	DATE	PROJECT	ihle engineering	Grubstr. 6-8 D-79279 Vörstetten	marc.ihle@ihle-engineering.de
ENG: *		DOC. REV. b3e31f4c34a7bc500a0742a3a152c5236e82cbd1 [Locally Modified]			
DSN: *		TITLE	*		
CHK: *		REFERENCE DOCUMENTS			
BOM:		SIZE DWG NO.			REV
ASSY DWG:		A3			
FAB DWG:		FILE NAME AC-Power-Monitor.SchDoc			
PCB DWG:		SHEET 1 OF 1			

Top side (Scale 2:1)



Bill Of Materials

Line #	Designator	Name	Quantity	Purpose
	C1	CC0603KRX7R9BB103	1	
	C2, C9, C19, C20, C22	CGA3E1X7R1E105K080AC	5	
	C3, C6, C8	06035C102KAT2A	3	
	C4, C5, C7, C24	CGA3E2X7R1H104K080AA	4	
	C10	10nF	1	
	C11, C12	MKY22W21003I00	2	
	C13, C14, C21	C0603C332J5GACTU	3	
	C15	180µF	1	
	C16, C17	C0603C105K3RACTU	2	
	C18	180µF	1	
	C23	885012206085	1	
	C25	C0603C474K4RACTU	1	
	D1	400W, 54VDC	1	
	FB1, FB2, FB3, FB4	1800Ohm	4	
	IC1	TMCS1127C1AQDVGR	1	
	IC2	AMC3330DWER	1	
	J1	691131710006	1	
	J3, J4	61300811821	2	
	M1	AC/DC-Buck-Module	1	
	MOD1	WIRL-COMB_2617011025000_1	1	
	P1, P3, P4, P5, P6, P9	61300211121	6	3.3V to ESP32, Bridge 5.0V to 3.3V, A1N to ESP32, A1P to ESP32, A0 to ESP32, 5.0V to ESP32
	P2, P7	61300311121	2	Power-Outputs, JTAG-Power Selector
	P8	61301421121	1	JTAG-Interface
	P10	61300411121	1	WLAN-BOOT-Interfaxce
	R1, R3	RK73H2BTDD1804F	2	
	R2	CRCW080510K0FKEA	1	
	R4	CRCW060347K0FKEA	1	
	R5	CRGCQ0805F1M0	1	
	R9, R10, R11	CRCW0603470KFKEA	3	
	R12	KTR03EZPF4702	1	
	R13	CPF0603B20KE1	1	
	R14	AC0603FR-071KL	1	
	R15, R16	CRCW06031K50FKEA	2	
	RV1	275VAC, 350VDC	1	
	U1	MCP1700T-3002E/TT	1	
	U2	MSP430I2041TRHBR	1	
	XF1	696106003002	1	

Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
	Surface Material	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.04mm		Signal	GTL
<i>Prepreg</i>		<i>0.07mm</i>	<i>PP-006</i>	<i>Dielectric</i>	
CF-004	N / GND	0.07mm		Signal	G1
		<i>0.32mm</i>	<i>FR-4</i>	<i>Dielectric</i>	
CF-004	L / VCC	0.07mm		Signal	G2
<i>Prepreg</i>		<i>0.07mm</i>	<i>PP-006</i>	<i>Dielectric</i>	
Copper	Bottom Layer	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 0.69mm

Bill of Materials

Project Title: Bill of Materials for BOM Document [AC-Power-Monitor BomDoc]
Project File Name: AC-Power-Monitor.PjPDR
Assembly Variant: None

Altium

2175 Salk Avenue
Suite 200
Carlsbad, CA 92008
USA

Part Number	Description	Designator	Manufacturer	Manufacturer Part Number	Supplier	Supplier Part Number	Supplier Unit Price	Quantity		
CAP-2100-03000	CAP CER 1000PF 50V X7R 0603	C1	Yageo	CC06030X7R9B8103	Mouser	603-CC06030X7R9B8103	\$0.09	1		
CMP-08246-0018	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C2, C22	TDK	CGA3E1X7R1E106K08AC	Mouser	810-CGA3E1X7R1E106K	\$0.10	2		
CMP-2006-04378	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	C3, C6	Nyccera AVX	9803C102KAT2A	Mouser	980-9803C102K	\$0.02	2		
CMP-2000-07545	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C4, C7	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	\$0.03	2		
CMP-2000-07545	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C5	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	\$0.03	1		
CMP-2006-04378	General Purpose Ceramic Capacitor, 0603, 1nF, 10%, X7R, 15%, 50V	C8	Nyccera AVX	9803C102KAT2A	Mouser	980-9803C102K	\$0.02	1		
CMP-08246-0018	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C9, C19, C20	TDK	CGA3E1X7R1E106K08AC	Mouser	810-CGA3E1X7R1E106K	\$0.10	8		
890304023006C3	WCAP-FTX2 Film Capacitor, THT, L13 x W5 x H11mm, 10nF, 275V	C10	Würth	890304023006C3	Würth	890304023006C3	\$0.37	1		
C-unipolar	CAP CER 3300PF 50V 5% NP0 0603	C11, C12	WIMA	MY2WV2200300005D	Mouser	955-MY2WV2200300005	\$0.70	2		
CMP-2006-03334	CAP CER 3300PF 50V 5% NP0 0603	C13, C14, C21	KEMET	C0603C332J50ACTU	Mouser	89-C0603C332J50	\$0.25	3		
875075155001	WCAP-PSLC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H11.7mm, 180uF, 6.3V, WCAP-PSLC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H11.7mm, 180uF, 18V	C15, C18	Würth Electronics	875075155001, 875075355001	NoParam	Display	NoParam	720-6488-1-ND	2	
CMP-2006-07713-1		C16, C17	KEMET	C0603C10K39ACTU	Mouser			89-C0603C10K39	\$0.23	2
CMP-03913-5391	88501220005 Multilayer Ceramic Capacitor, Wrap-Cap Series, 2200PF, 10%, X7R, 50 V, 0603 1808 Metric RoHS Compliant: Yes.	C23	Würth Electronics	8.885012E-11	Mouser			710-88501220005	\$0.09	1
CMP-2000-07545	Cap Ceramic 100nF 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C24	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	\$0.03	1		
CMP-2006-03235	CAP CER 0.47uF 16V X7R 0603	C25	KEMET	C0603C474K6RACTU	Mouser	89-C0603C474K6R	\$0.11	1		
824901541	WE-TVSP SMT Power TVS Diode, DO-21AC, Bidirectional, 400W, 54V/DC	D1	Würth Electronics	824901541				710-824901541	\$0.26	1
74269244180	WE-TMSB Tiny Multilayer Suppression Bead, Size 0402, 1800ohm, 0.21A	F1, FB2, FB3, FB4	Würth Electronics	74269244180	Dikey			710-74269244180CT-ND	\$0.10	4
AMC330DWER	Integrated Circuit	C-2	Texas Instruments	AMC330DWER	Mouser	595-AMC330DWER	\$0.50	1		
6.91132E+11	WR-TB Series 1317 Horizontal Entry, THT, pitch 6.91132E+11 5mm, 09g	J1	Würth Electronics	6.91132E+11	Mouser			710-691131710006	\$3.73	1
61300811821	WR-PHD Socket Header, THT, Vertical, pitch 2.54mm, 1 Row, 8P	J3, 14	Würth Electronics	61300811821	Mouser			710-61300811821	\$0.38	2
AC/DC-Block-Module	Diaphano-1 - Combined 2.4 GHz Wi-Fi and Bluetooth®	M1	Traco Power	TMP03-105	Mouser			495-TMP03-105	\$10.88	1
WIRL-COMB 2631E 5.0 Module		MOD1	Würth Elektronik	2.61701E+12	Dikey			710-261701102000008-ND	\$8.70	1
61300211121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 3P	P1, P3, P4, P5, P6, P9	Würth Electronics	61300211121	Mouser			710-61300211121	\$0.12	6
61300311121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 3P	P2, P7	Würth Electronics	61300311121	Mouser			710-61300311121	\$0.12	2
61301421121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 2 Row, 14P	P8	Würth Electronics	61301421121	Mouser			710-61301421121	\$0.88	1
61300411121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 4P	P10	Würth Electronics	61300411121	Mouser			710-61300411121	\$0.17	1
CMP-2003-01283	1.8MO ±1% 0.25W Chip Resistor 1206 Thick Film AEC-Q200 Qualified	R1, R3	NDA Speer	R173H28T01804F	Mouser			660-R173H28T01804F	\$0.09	2
CMP-2001-04809	RES Thick Film, 10kΩ, 1%, 1/10W, 100ppm/°C, 0805	R2	Vishay	CRCW080510K9F6EA	Dikey			541-1010C27-ND	\$0.09	1
CMP-2000-03379-1/10W 0603	RES SMD 47K OHM 1% 1/10W 0603	R4	Vishay	CRCW0603470F6EA	Mouser			71-CRCW0603-47K-63	\$0.14	1
CMP-03211-0034	CRGQ0805 1MO 0.125W 100ppm/°C 1% 180V	R5	TE Connectivity	CRGQ0805F1M0	Arrow Electronics	CRGQ0805F1M0	\$0.00	1		
CMP-2000-03370-0	RES Thick Film, 470kΩ, 1%, 0.1W, 100ppm/°C, 0603	R9, R10, R11	Vishay	CRCW0603470F6EA	Newark			5N9555	\$0.05	3
CMP-08839-0306	Automotive Paper T/R RES SMD 20KΩ 0.1% 1/8W	R12	Rohm	KTR032ZF4702	Mouser			755-KTR032ZF4702	\$0.15	1
CMP-00211-0063	RES 1kΩ 0.1W 0603	R13	TE Connectivity	399-0603B20K61	Arrow Electronics	399-0603B20K61	\$0.10	1		
CMP-2000-04809	RES 1kΩ 0.1W 0603	R14	Yageo	AC0603PR-0714	Dikey			311-JALC27-ND	\$0.09	1
CMP-2000-02995-0	RES Thick Film, 1.5kΩ, 1%, 0.1W, 100ppm/°C, 0603	R15, R16	Vishay	CRCW0603150F6EA	Avnet			61481391	\$0.03	2
820412711	WE-VD Disk Varistor, THT, 27V/4A, 35V/DC, 150pF Low Quiescent Current LDO, 3-Pin SOT-23, Extended Temperature, Tape and Reel	RV1	Würth Electronics	820412711	Mouser			710-820412711	\$0.52	1
CMP-0136-0078	IC MCU 16BIT 32KB FLASH 3V/0VFN	U1	Microchip	MCU17001-3002E/TT	Arrow Electronics	MCU17001-3002E/TT	\$0.43	1		
CMP-04869-0003	WE-FSA Fuseholder THT	U2	Texas Instruments	MSF4302041TRH8R	Mouser			395-MSF4302041TRH8R	\$2.87	1
6.9E10E+11	Blocks PCB Clip Cover	XF1	Würth Electronics	6.9E10E+11	Mouser			710-69E10E00002	\$0.53	1
								\$43.71	64	

Approved	Notes
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